

Page 9

Definition 1.1

The *mean* of a sample

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Page 10

Definition 1.2

The *variance* of a sample

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2$$

Page 10

Definition 1.3

The *standard deviation* of a sample

$$s = \sqrt{s^2}$$

Page 30

Definition 2.6

The *probability* of event A in S

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

Page 43

Definition 2.7

An ordered arrangement of r distinct objects is called a *permutation*

$$P_r^n = n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!}$$

Page 44

Theorem 2.3

The number of ways of partitioning n distinct objects into k distinct groups

$$N = \frac{n!}{n_1!n_2!\dots n_k!}$$

Page 36

Definition 2.8

The number of *combinations* of n objects taken r at a time is the number of subsets, each of size r , that can be formed from the n objects

$$C_r^n = \frac{n!}{r!(n-r)!}$$

Page 52

Definition 2.9

The *conditional probability* of an event A, given that an event B has occurred, is equal to

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Page 53

Definition 2.10

Two events A and B are said to be independent if any one of the following holds, otherwise, the events are said to be dependent

$$P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

$$P(A \cap B) = P(A)P(B)$$

Page 57

Theorem 2.8

The Multiplicative Law of Probability The probability of the intersection of two events A and B is

$$P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$$

If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

Page 58

Theorem 2.6

The Additive Law of Probability The probability of the union of two events A and B is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive events ($P(A \cap B)=0$) then,

$$P(A \cup B) = P(A) + P(B)$$

Page 59

Theorem 2.7

If A is an event, then

$$P(A) = 1 - P(\bar{A})$$

Page 70

Definition 2.11

The *law of total probability*

$$P(A) = \sum_{i=1}^k P(A|B_i)P(B_i)$$

Page 71

Theorem 2.9

The *Bayes Rule*

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{i=1}^k P(A|B_i)P(B_i)}$$

Page 88

Definition 3.3

The *probability distribution* for a discrete variable Y can be represented by a formula, a table, or a graph that provides

$$p(y) = P(Y = y) \text{ for all } y$$

Page 91

Definition 3.4

the *expected value* of Y , $E(Y)$, is defined to be

$$E(Y) = \sum_y yp(y)$$

Page 93

Theorem 3.2

Let Y be a discrete random variable with probability function $p(y)$ and $g(Y)$ be a real-valued function of Y . Then the expected value of $g(Y)$ is given by

$$E[g(Y)] = \sum_{all\ y} g(y)p(y)$$

Page 93

Definition 3.5

If Y is a random variable with mean $E(Y) = \mu$, the *variance* of a random variable Y is defined to be

$$V(Y) = E[(Y - \mu)^2]$$

Page 103

Definition 3.7

A random variable Y is said to have a *binomial distribution* based on n trials with success probability p if and only if

$$p(y) = nCy p^y q^{n-y}$$

Page 107

Theorem 3.7

Let Y be a binomial random variable based on n trials and success probability p . Then

$$\mu = E(Y) = np \quad \text{And} \quad \sigma^2 = V(Y) = npq$$

Theorem 3.8

If Y is a random variable with a *geometric distribution*

$$E(Y) = \sum_{y=1}^{\infty} yq^{y-1}p$$

$$\mu = E(Y) = \frac{1}{p} \quad \text{And} \quad \sigma^2 = V(Y) = \frac{1-p}{p^2}$$